

GENERICS

```
// class
class Box<T> {}

// variable
Box<String> b = new Box<String>();
// method
public <T> void doStuff(T value) {}
```

JAVA BEING STUPID

```
T obj = new T(); // error
obj instanceof T; // error
T heh = (T) "asdf"; // no typechecking
var s = Stack<int>(); // no primitive types
private static T x; // static no worky
Node<Number> s = new Node<Integer>; // error
class IThrowAnErrorBecauseWhyNot {
    void m(List<String> list) {}
    void m(List<Integer> list) {}
} // error because of type erasure & identifiability
```

ITERATOR

```
for (String s : stringList) {} // ==
for (Iterator<String> i = stringList.iterator();
    i.hasNext();) {
    String s = i.next();
}
```

ITERABLE

```
interface Iterable<T> {
    Iterator<T> iterator();
}

interface Iterator<E> {
    boolean hasNext();
    E next();
}
```

COMPARABLE

TODO:

Type signature?

```
static <T extends Comparable> T doStuff(T a, T b) {
    // compareTo is possible with all types
    return a.compareTo("b") > 0 ? a : b;
}

static <T extends Comparable<T>> T doStuff(T a, T b)
{
    // compareTo is possible only with T
    return a.compareTo(b) > 0 ? a : b;
}
```

EXAMPLE

```
class Graphic {
    public void draw() {}
}

class Rectangle extends Graphic { }
class Circle extends Graphic { }
class GraphicStack<T extends Graphic>
    extends Stack<T> {
    public void drawAll() {
        for (T item : this) item.draw();
    }
}
```

BYTECODE

TODO:

TYPE ERASURE

– Introduced because of "bAcKwArDs CoMpAtAbIlItY"

– No type information at runtime (Non-Reifiable Type)

```
class MyStack<T> {
    Entry<T> top;
    int push(T value) {}
}

// becomes
class MyStack {
    Entry top;
    int push(Object value) {}
}

/* with bounds */
class MyStack<T extends Comparable<T>> {
    Entry<T> top;
    int push(T value) {}
}

// becomes
class MyStack {
```

```
Entry top;
int push(Comparable value) {}
}

/* what happens at/before runtime */
MyStack<String> s = new MyStack<String>();
s.push("Eh");
String x = s.pop();
// becomes
MyStack s = new MyStack();
s.push("Eh");
String x = (String) s.pop();
```

WILDCARDS

```
void printGs(List<Graphic> gs) {}
List<Graphic> gs1 = new ArrayList<>();
printGs(gs1); // ok
List<Circle> gs2 = new ArrayList<>();
printGs(gs2); // error
/* solution */
void printGs(List<? extends Graphic> gs) {}
```

GENERIC VARIANCE

|                | Type           | Compatible Type-Args | R | W |
|----------------|----------------|----------------------|---|---|
| Invariance     | C<T>           | T                    | ✓ | ✓ |
| Covariance     | C<? extends T> | T and sub-types      | ✓ | ✗ |
| Contravariance | C<? super T>   | T and basis-types    | ✗ | ✓ |
| Bivariance     | C<?>           | All                  | ✗ | ✗ |

INVARIANCE

```
static <T> void move(Stack<T>, from, Stack<T> to) {
    while(!from.isEmpty()) to.push(from.pop());
}

var rectS = new Stack<Rectangle>();
var graphS = new Stack<Graphic>();
graphS.push(new Rectangle()); // ok
move(rectS, graphS); // error
Stack<Object> objS = graphS; // error
/* anotherone */
String[] strA = new String[10];
Object[] objA = strA; // ok
objA[0] = Integer.valueOf(2); // runtime err
```

TODO:

```
// compiles?
List<Graphic> list = new ArrayList<Rectangle>();
// compiles?
Rectangle[] rectangleStack = new Rectangle[10];
Graphic[] graphicStack = new Graphic[10];
graphicStack = rectangleStack;
// compiles?
Stack<Rectangle> rectangleStack = new Stack<>();
Stack<Graphic> graphicStack = new Stack<>();
graphicStack = rectangleStack;
// compiles?
Stack<Rectangle> rectangleStack = new Stack<>();
Stack<Graphic> graphicStack = new Stack<>();
for (Rectangle r : rectangleStack) {
    graphicStack.push(r);
}
```

COVARIANCE

```
public class Stack<E> {
    public void pushAll(Iterable<? extends E> src) {}
}

Stack<? extends Graphic> graphS;
graphS = new Stack<Rectangle>(); // ok
graphS = new Stack<Circle>(); // ok
graphS = new Stack<Object>(); // error (duh)
/* read only */
Stack<? extends Graphic> stack = new
    Stack<Rectangle>();
stack.push(new Graphic()); // error
stack.push(new Rectangle()); // error
stack.push(new Triangle()); // error
```

CONTRAVARIANCE

```
static <T> void addToC(List<? super T> list, T e) {
    list.add(e);
}

addToC(new ArrayList<Number>(), 3); // ok
addToC(new ArrayList<Object>(), 3); // ok
/* shenanigans */
```

```
Stack<? super Graphic> s = new Stack<Object>();
s.add(new Graphic()); // ok
s.add(new Rectangle()); // ok
Graphic g = s.pop(); // error
Object g = s.pop(); // ok
```

BIVARIANCE

```
static void printList(List<?> list) {
    for (Object elem : list)
        System.out.print(elem + " ");
}

/* more shenanigans */
static void appendNewObject(List<?> list) {
    list.add(new Object()); // error
}
```

PRODUCER / CONSUMER

TODO:

MISC

<T extends Comparable & Collection> // multiple type bounds

GENERIC STREAM TOMFOOLERY

```
List<? extends Media> mediaList;
public <S extends T> List<S>
    filterBySubtype(Class<S> subtype) {
    return mediaList.stream()
        .filter(subtype::isInstance)
        .map(subtype::cast)
        .collect(Collectors.toList());
}
```

ANNOTATIONS

– Metadata

– Not actually part of the code

TODO:

slides 10

– @Override

– @Test

– @Json...

DEFINING

```
@Target(ElementType.TYPE)
@Retention(RetentionPolicy.RUNTIME)
public @interface Author {
    String name();
    String date() default "N/A";
}
```

```
@Author(name = "UwU", date = "idon't validate input")
public class WeeOoo {}
```

REFLECTION

– Information of Class (private and public): Methods, Attributes, Parent class, Implemented interfaces

– Calling methods possible

– Usages: Debugger, Serialization, Frameworks, Remote Procedure Call, ORM

// all of these `throws SecurityException`

```
public Field[] getDeclaredFields()
public Method[] getDeclaredMethods()
public Constructor<?>[] getDeclaredConstructors()
```

```
System.out.println(dump(new Student("a", "b"), 0));
// {
//     firstName = a
//     lastName = b
// }
Class c = "s".getClass(); // java.lang.String
```

TODO:

Diagram (slides 26)

METHOD

// @Target(ElementType.METHOD)

```
public String getName()
public boolean isAnnotationPresent(
    Class<? extends Annotation> annotationClass)
public Object invoke(Object obj, Object... args)
    throws IllegalAccessException,
```

```
IllegalArgumentException,
InvocationTargetException

public class Profiler {
    public static void main(String[] args) {
        var testFunctions = new ProfileFunctions();
        int[] array = {5, 2, 8, 1, 93, 3, 33, 1, 333};
        var methods = ProfileFunctions.class
            .getDeclaredMethods();
        for (var m : methods) {
            if(m.isAnnotationPresent(Profile.class)) {
                m.setAccessible(true); // best practice or
                sth, idk, i write code in actually good languages
                Profiler.profileMethod(testFunctions, m,
                    new Object[] {array});
            }
        }
    }
}
```

CLASS

TODO:

// @Target(ElementType.TYPE)

```
getDeclaredConstructor().newInstance()
...
// @Target(ElementType.FIELD)

VALIDATION



```
@Min(value = 18, message = "Age must be ≥ 18")
@Max(value = 99, message = "Age must be ≤ 99")
private int age;
@NotNull(message = "Name cannot be null")
private String name;
```



ALGORITHMS



– Brute-force



– Greedy



– Take best option each step (optimize locally)



– Fast



– Backtracking



– Trial and error



– Best overall option



– Higher compute costs



– Divide and conquer



– Binary search



– Dynamic programming



– Recursion optimization?



– Tabulation



BACKTRACKING



TODO:



slides 87



BIG O NOTATION



– Worst case scenario



– Atomic operations = constant time



– Runtime measured as sum of primitive operations



Notation



$$\underset{\text{Algorithm}}{f} \left( \underset{\text{Size of problem}}{n} \right) = \text{Number of steps}$$



$g$  = complexity class



TODO:



diagram  $f \leq cg$


```

$f(n)$  is  $O(g(n))$  if  $c > 0 \wedge n_0 \geq 1$

$f(n) \leq cg(n)$  for  $n \geq n_0$

$f$  in order  $g$ :

Let  $f, g: \mathbb{N} \rightarrow \mathbb{N}$

$n_0, c \in \mathbb{N} \wedge \forall n \geq n_0: f(n) \leq c \cdot g(n)$

$\Rightarrow f \in O(g) \Leftrightarrow f = O(g)$

Big O

$$\widetilde{O} \left( \underset{\text{Size of problem}}{n} \right) = \text{Complexity class}$$

RULES

If  $f(n)$  is polynomial of degree  $d$ , then  $f(n) \in O(n^d)$

– ignore lower powers

– ignore constants

Use most optimal function (lowest power)

–  $2n$  is  $O(n)$  better than  $2n$  is  $O(n^2)$

Simplify as far as possible

–  $3n + 5$  is  $O(n)$  instead of  $3n + 5$  is  $O(3n)$

COMPLEXITY CLASSES

| Name        | Class      | Example                    |
|-------------|------------|----------------------------|
| Constant    | 1          | Array read                 |
| Logarithmic | $\log n$   | Binary search              |
| Linear      | $n$        | Linear search              |
| Log-linear  | $n \log n$ | Merge+Quick sort           |
| Quadratic   | $n^2$      | Bubble+Insertion sort      |
| Cubic       | $n^3$      | Solve equation             |
| Polynomial  | $n^k$      | Various algorithms         |
| Exponential | $2^n$      | Calculate all subsets      |
| Factorial   | $n!$       | Calculate all permutations |

TODO:

Counting primitive operations (slides 55)

SORTING ALGORITHMS

SELECTION SORT

– Search for smallest elem in unsorted part and move in front

- Less mutations in array
- Same performance even if list almost sorted

```
public static void selectionSort(int[] a) {
    int n = a.length;
    for (int i = 0; i < n - 1; i++) {
        int minimum = i;
        for (int j = i + 1; j < n; j++)
            if (a[j] < a[minimum])
                minimum = j;
        swap(a, i, minimum);
    }
}
```

$$(n-1) + (n-2) + \dots + 1 = \sum_{i=1}^{n-1} i = \frac{n(n-1)}{2} = \frac{n^2-n}{2} \sim O(n^2)$$

- INSERTION SORT**
- Take elem and put into correct place in sorted part
  - Good for already partially sorted arrays

```
public static void insertionSort(int[] a) {
    int n = a.length;
    for (int i = 1; i < n; i++)
        for (int j = i; j > 0 && a[j] < a[j - 1]; j--)
            swap(a, j, j - 1);
}
```

$$1 + \dots + (n-1) + n = \sum_{i=1}^n i = \frac{n(n+1)}{2} = \frac{n^2+n}{2} \sim O(n^2)$$

- BUBBLE SORT**
- Compare neighboring elems and swap if needed

```
public static void bubbleSort(int[] a) {
    for (n = a.length; n > 1; n--)
        for (i = 0; i < n - 1; i++)
            if (a[i] > a[i + 1])
                swap(a, i, i + 1);
}
```

$$(n-1) + (n-2) + \dots + 1 = \sum_{i=1}^{n-1} i = \frac{n(n-1)}{2} = \frac{n^2-n}{2} \sim O(n^2)$$

```
public static void countingSort(int[] a) {
    int k = Arrays.stream(arr).max().getAsInt();
    int[] c = new int[k + 1];
    for (int i : a) c[i] += 1;
    int pos = 0;
    for (int i = 0; i < k; i++)
        for (int j = 0; j < c[i]; j++) {
            a[pos] = i;
            pos++;
        }
}
```

$$n + n + k + n = 3n + k \sim O(n + k)$$

- SHELL SORT**
- Sort pairs of far away elems
  - Sort the partially sorted array through insertion sort

```
public static void shellSort(int[] a) {
    int n = a.length;
    int h = 1;
    while (h < n/3) h = 3 * h + 1;
    while (h >= 1) {
        for (int i = h; i < n; i++)
            for (int j = i; j >= h && a[j] < a[j - h]; j = j - h) swap(a, j, j - h);
        h /= 3;
    }
}
```

**BINARY SEARCH**

```
public static <T extends Comparable<T>> boolean bs(
    List<T> data, T target, int low, int high) {
    if (low > high) return false;
    int pivot = low + ((high - low) / 2);
    if (target.equals(data.get(pivot)))
        return true;
    else if (target.compareTo(data.get(pivot)) < 0)
        return searchBinary(data, target, low, pivot - 1);
    else
        return searchBinary(data, target, pivot + 1, high);
}
```

```
return searchBinary(data, target, pivot + 1, high);
}
```

$$O(\log n)$$

**RECURSION**

When a function calls itself

**LINEAR RECURSION**

Recursive call starts **at most one** further recursive call

**TODO:**

recursive → iterative (slides 61)

**TAIL RECURSION**

Linear recursion where the recursive call is **last**

**TODO:**

recursive → iterative (slides 69 – nice)

**BINARY RECURSION**

All non-terminal calls have **two** recursive calls

**TODO:**

example (slides 76,77,78)

**TODO:**

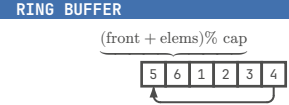
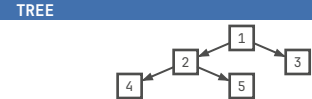
$O(n)$  (slides 84)

**DATA STRUCTURES**

ADT: Abstract Data Type (e.g. Interface)

**TODO:**

diagram



**PRIORITY QUEUE**

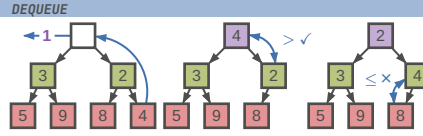
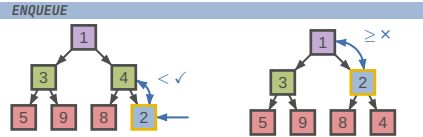
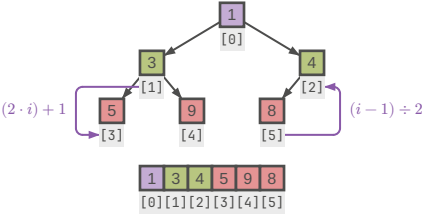
| Method    | Unsorted List | Sorted List |
|-----------|---------------|-------------|
| insert    | $O(1)$        | $O(n)$      |
| min       | $O(n)$        | $O(1)$      |
| removeMin | $O(n)$        | $O(1)$      |

**HEAP**

- Binary tree
- Layers 0, ...,  $h-1$  fully populated, layer  $h$  filled from left
- Operations always lowest RHS so that tree stays consistent

**Min-Heap:** Keys of nodes are **smaller** or equal than children's

**Max-Heap:** Keys of nodes are **larger** or equal than children's



**HEAP SORT**

Dequeue elems from head into list, resorting tree on each iter

**TODO:**